Development of an accessible gait analysis method based on Xsens sensors for incomplete spinal cord injury patients

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Introduction

Wearable sensor-based motion-capture systems, such as Xsens, are a promising alternative to optical systems for implementing gait analysis in rehabilitation centers, since the latter requires special infrastructure that is not feasible in most clinical settings. However, the Xsens calibration procedures rely on anatomically neutral positions (static calibration) and movements (moving calibration), which spinal cord injury (SCI) patients commonly struggle with or cannot perform.

Since no studies were found addressing this limitation, our objective is to investigate whether an alternative and accessible calibration method for the full-body Xsens-Awinda system can be developed and implemented in clinical environments.

Methods

We developed a calibration workflow that combines Xsens recordings with OpenSense (OpenSim) [1,2]. This method requires doing an Xsens recording while two cameras simultaneously capture the frontal and sagittal view of the subject, while they hold a comfortable position for ≥3 s, with assistive devices if needed. The photographs are then used to align a scaled musculoskeletal model with the subject's pose and, uning the Xsens measurements, IMUs are rigidly attached to the model's body segments in OpenSim, to replicate experimental placement. Using OpenSense inverse kinematics, joint angles were computed for subsequent trials by minimizing the error between IMU recordings and body segment orientations.

Data were available from 4 healthy and 3 SCI subjects walking at a comfortable speed. Both groups performed a static calibration that Xsens classified as "good", and healthy subjects also performed a "good" moving calibration. For each walking trial, we had the angles from Xsens and those obtained with the developed method. We computed the inter-trial (between two trials with the same calibration) and inter-method (between Xsens' and OpenSim's calibration of the same trial) variability as the root mean squared error (RMSE) normalized to range of motion (ROM) for hip extension, knee extension, and ankle dorsiflexion.

Results

The inter-trial variability of SCI subjects, both for Xsens and OpenSim calibrations, is always higher than that of healthy subjects (288% higher, on average). Moreover, the inter-trial variability is very similar when computed with Xsens or with OpenSim (less than 0.06% of ROM of difference for healthy subjects and less than 0.38% of ROM for SCI subjects). The inter-method variability is higher for a static than for a moving calibration in healthy subjects (average RMSE was 2.71 and 2.09% of ROM); and higher for SCI subjects than for healthy subjects when performing static calibration (3.27 versus 2.71% of ROM). (Fig 1)

Assuming that the difference between angles from an Xsens "good" moving calibration and our method is equivalent to the difference between actual movements and our computed angles [3], static calibrations would lead to larger errors than moving calibrations. The error increase from healthy to SCI subjects suggests that the static calibration is even less accurate for SCI subjects, despite both being classified as "good" by Xsens, highlighting the need for alternative calibration methods.

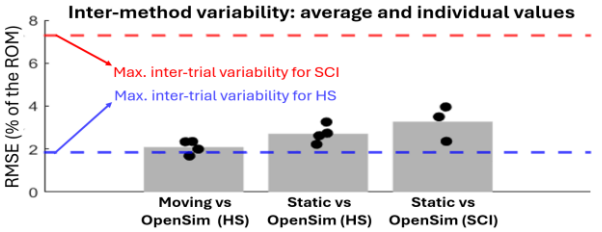


Figure 1: Inter-method variability for the measurements obtained with moving and static calibration in Xsens for healthy (HS) and SCI subjects, with dots representing individual values and bars showing the averages. Dashed lines show the maximum averaged inter-trial variability for SCI (red) and healthy subjects (blue), for any method.

Conclusions

The proposed method seems able to compute joint angles from Xsens measurements for healthy subjects and a moving calibration, and seems to offer a feasible alternative when the subjects are unable to get a reliable calibration in Xsens. However, further validation with larger cohorts, and comparisons with optical motion capture would be needed to ensure the computed angles do not depend on the original Xsens calibration quality.

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Referencias

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